

FACULTY OF INFORMATICS & DESIGN

Diploma in Information & Communication Technology

INFORMATION SYSTEMS II: ITS262S

SUBJECT GUIDE:	2021
COURSE CODE:	ITS262S
NQF LEVEL:	6
PREREQUISITE SUBJECTS:	PREREQUISITE SUBJECTS: n/a

Table of Contents

1. INTRODUCTION	4
2. GENERAL	4
2.1 Contact Information	4
2.2 Class Times and Venues.....	Error! Bookmark not defined.
2.3 Learner Management System (LMS)	5
3. STUDY MATERIALS	5
3.1 Prescribed Textbook	5
3.2 Recommended Readings	5
3.3 Websites	5
4. MODULE CREDITS.....	5
4.1 Planning of Time Allocation for Learning Activities.....	6
5. ETHICAL CONDUCT.....	6
5.1 Roles and Responsibilities	6
5.2 Class attendance and participation	6
5.4 Plagiarism	6
6. ASSESSMENT.....	7
6.1 Assessment policy and regulations	7
6.2 Assessment Opportunities: Administration	7
7. SUBJECT SPECIFICATIONS	8
7.1 Purpose of the Subject	8
7.2 Subject Structure	8
7.2.1 Subject Outcomes.....	8
7.2.2 Subject Breakdown Schedule	9
7.3 Articulation with other subjects in the programme.....	13

7.4	Learning presumed to be in place	13
7.5	Critical Cross-field Outcomes	13
8.	STUDY UNITS/STUDY THEMES.....	13
8.1	Study Units.....	13
8.2	Assessment Opportunities and Assessment strategy	13
8.3	Subject Outcomes and Associated Assessment Criteria	13
8.4	Self-study Activities	13
9.	STUDENT ACKNOWLEDGEMENT	14

ORGANISATIONAL COMPONENT

1. INTRODUCTION

Welcome to the 2nd year of the Diploma in Information and Communication Technology in Application Development. Information Systems II is the application development specialisation area.

The purpose of this subject is to provide students with the fundamental knowledge of designing simple databases by means of entity relationship diagrams and the implementation of these designs in a relational database management system. Effective databases are essential tools in the contemporary business environment, and the ability to store, modify, extract and search for information through databases is a valuable skill in this environment. Leveraging project based learning techniques, students will create and work with problems which challenge them to design, implement, and demonstrate a database solution for a business or organization

2. GENERAL

2.1 Contact Information

DESIGNATION	NAME	LOCATION	TELEPHONE NUMBER	E-MAIL ADDRESS
Subject Co-ordinator	F, Allie	Engineering 2.47	(021) 460 7203	allief@cput.ac.za
Lecturers	F, Allie	Engineering 2.47	(021) 460-7203	allief@cput.ac.za
	K.Swart	Engineering		swartKS@cput.ac.za
Domain Coordinator	K. Naidoo	Engineering 2.05	(021) 460-3348	naidook@cput.ac.za
Head of Department	B. Kabaso	Engineering 2.30	(021) 460-3713	kabasob@cput.ac.za
Administrative Assistant	D. O'Brien	Engineering 2.49	(021) 460-3685	pillaydl@cput.ac.za
Support Technician	S. Salie	Engineering 1.4	(021) 460-4209	saliesh@cput.ac.za
Support Technician	R. Pietersen	Engineering 1.4	(021) 460-4209	pietersenr@cput.ac.za

You are most welcome to speak to your lecturer personally, please email for an appointment.

If you have any queries relating to marks that have not been resolved or any other issue such as missing an exam, please do not email me but use the IT Admin system. Please indicate the subject code and lecturer surname so that the ticket can be processed quickly.

This is where you register the complaint / issue:

To open or create a new ticket/query

<https://enquiries.cput.ac.za/>

As published on <http://myclassroom.cput.ac.za> and notice boards. The subject consists of two theory periods per week during term 1.

2.3 Learner Management System (LMS)

<https://myclassroom.cput.ac.za/>: Blackboard used for general communications and announcements, posting of course material, assignment and announcements. It is the responsibility of the learner to check Blackboard regularly.

3. STUDY MATERIALS

3.1 Prescribed Textbook

Coronel / Morris. Database systems: Design, implementation and management. 12th ed. Boston: Cengage Learning. ISBN: 978-1-305-62748-2

3.2 Recommended Readings

3.3 Websites immediately

- www.bookboon.com
- <http://eclassroom.cput.ac.za> : Blackboard/WebCT used for general communications and announcements.

4. MODULE CREDITS

This module has 0.083 credits on NQF level 6. Please refer to the Faculty prospectus on

<http://www.cput.ac.za/academic/faculties/informaticsdesign/prospectus> for further information.

4.1 Planning of Time Allocation for Learning Activities

There are 100 notional hours allocated to this module. These notional hours comprise of 60 hours (30%) for contact time and the remainder will be self-study and other tasks, assignments, and tests.

METHOD	% OF YOUR LEARNING TIME
Lectures	30%
Independent self-study of standard texts and references and specially prepared materials (study guides, books, journals, using online tutorials etc.)	20%
Independent self-study, time needed to prepare for assessment (e.g. tests/exams)	25%
Student learning groups (working with your peers for group assignments)	25%

5. ETHICAL CONDUCT

5.1 Roles and Responsibilities

Kindly refer to CPUT's Policy Docs and Student Guidelines:

https://www.cput.ac.za/storage/study_at_cput/Student_Rules%20Regulations_2018.pdf

5.2 Class attendance and participation

Kindly refer to page 18 of CPUT's General Handbook Academic and Student Rules and Regulations:

https://www.cput.ac.za/storage/study_at_cput/Student_Rules%20Regulations_2018.pdf

5.3 Grievance procedures

Kindly refer to page 61 of CPUT's General Handbook Academic and Student Rules and Regulations:

https://www.cput.ac.za/storage/study_at_cput/Student_Rules%20Regulations_2018.pdf

5.4 Plagiarism

Kindly refer to page 30 of CPUT's General Handbook Academic and Student Rules and Regulations:

https://www.cput.ac.za/storage/study_at_cput/Student_Rules%20Regulations_2018.pdf

*NB: An amended version of the General Handbook will be emailed to you during the first term.

6. ASSESSMENT

6.1 Assessment policy and regulations

This subject has four assessments. Assessments are a combination of Blackboard online tests, iLearning quizzes and assignments.

Test queries: Last day for any queries regarding a test is one week after the test was discussed in class. Thereafter the test mark cannot be changed.

Absence from a test: If you are absent from a test, a valid medical certificate must be given to your lecturer (via the Secretary) within one week after the test was written. You must also inform your lecturer on the day of the test that you are ill.

Due dates must be strictly adhered to. NO late submissions are allowed for any subject in this Diploma.

Assignments are administered via Blackboard.

6.2 Assessment Opportunities: Administration

There will be one test per term. There will be a sick test in June for the first semester formal assessments and a sick test in November for the second semester formal assessments.

ASSESSMENT BREAKDOWN AND WEIGHTS							
ASSESSMENT CRITERIA	1 ST TERM	2 ND TERM		3 RD TERM		4 TH TERM	TOTALS
Nature of assessment	Test(s) / Assignment / Tutorials	Written Test	Tests/ Tutorials	Written Test	Tests/ Tutorials	Assignment	
Will assessment be included in this term?	No	Yes	No	Yes	No	No	100%
Total assessment weight for the term	10%	30%	5%	30%	5%	20%	
Will the assessment be moderated?	No	Yes	No	Yes	No	No	

STUDY COMPONENT

7. SUBJECT SPECIFICATIONS

7.1 Purpose of the Subject

This is an Introduction to database design and provides you with the fundamental knowledge of designing simple databases by means of entity relationship diagrams for implementation in a relational database environment. This course aims to provide practical, hands-on, engaging activities. Leveraging project-based learning techniques, students will create and work with problems which challenge them to design, implement, and demonstrate a database solution for a business or organization.

7.2 Subject Structure

7.2.1 Subject Outcomes

SUBJECT OUTCOMES	DESCRIPTION
SO1	Understanding of database concepts, database models.
SO2	Demonstrate the ability to understand and design relational databases.
SO3	Understand and demonstrate the ability to be able to model an application's data requirements using conceptual modelling tools like Entity Relationship diagrams and design database schemas based on the conceptual model.
SO4	Apply normalisation of database tables up to 3 rd normal form.
SO5	Have a good understanding of transaction management.

7.2.2 Subject Breakdown Schedule

*

TERM 1				
STUDY UNIT 1: DATABASE CONCEPTS				
SUBJECT OUTCOMES	THEMES/SPECIFIC OUTCOMES	CHAPTER	ACADEMIC SCHEDULE	ASSESSMENT DATES
SO1	UNDERSTANDING OF DATABASE CONCEPTS			
	Introduction	1	01 March	
	Understand and explain basic database terms and concepts		03 March	
	Understand the importance of database design			
	Understand the objectives of a database management system			
	Understand the architecture of a database management system (DBMS)			
	Understand why data models are important	2	08 March	
	Understand basic data-modeling building blocks		11 March	
	Understand the evolution of the hierarchical, network and relational model		15 March	
	Understand business rules			
	Understand the logical view of data			
SO2	Test /Tut			
	INTRODUCTION TO RELATIONAL DATABASES			
	Understand the logical view of data	3	22-March	
	Understand the relational model's basic component, relations and keys		23 March	
	Understand the data integrity rules		29 March	
	The data dictionary			
	Understand how data redundancy and indexes is handled			
	Test /Tut			

RECESS : 1 April – 9 April

- 27 April: Freedom Day

STUDY UNIT 3 DESIGN CONCEPTS

SUBJECT OUTCOMES	THEMES/SPECIFIC OUTCOMES	CHAPTER	ACADEMIC SCHEDULE	ASSESSMENT DATES
SO3	ENTITY RELATIONSHIP MODELING			
	Understand the main characteristics of entity relationship components. Understand how relationships between entities are defined, refined, and incorporated into the database design process.	4	12 April 2021 19 April 2021	
SO4	Understand how ERD components affect database design and implementation. Explain the characteristics of good primary keys and how to select them. Apply the business rules to create a Crow's foot ERD Test/Tut	4	26 April 2021 10 May 2021	
	Understand and apply the definitions of the normalization process (0NF, 1NF, 2NF, &3NF)	6	24 May 2021	
	Test/Tut			
	TEST Week			
	Project and sick test			
	Sick test week			

TERM 3 19 July -

STUDY UNIT: Normalisation and transaction management

SUBJECT OUTCOMES		CHAPTER	ACADEMIC SCHEDULE	ASSESSMENT DATES
SO4 SO5	Normalisation of Database tables			
	What normalization is and the role it plays in the database design process. How to convert normal forms up to 3rd normal form. Convert ERDs to a Relational Database Identify and eliminate relationship anomalies. Normalisation and ERD modeling used concurrently to produce a good database design.	6	19 July 2021 26 July 2021 02 August 2021	
	TRANSACTION MANAGEMENT			
	Define database transactions and their properties Understand what concurrency control is and what role it plays in maintaining the database's integrity Explain what locking methods are and how they work Determine how stamping methods are used for concurrency control Understand how optimistic methods are used for concurrency control Explain how database recovery management is used to maintain database integrity Test/tut/project	10	09 August 2021 16 August 2021	
	Test Week 1			
	Test Week 2			

•

TERM 4

STUDY UNIT: ASSIGNMENT				
SUBJECT OUTCOMES	THEMES/SPECIFIC OUTCOMES	CHAPTER	ACADEMIC SCHEDULE	ASSESSMENT DATES
	ASSIGNMENT			
	ASSIGNMENT		27 Sept 2021	
	ASSIGNMENT		04 October 2021	
	Consultation		11 October 2021	
	Consultation		18 October 2021	
	Consultation		25 October 2021	
RECESS				

7.3 Articulation with other subjects in the programme

This subject is linked to concepts taught in Information Systems II.

7.4 Learning presumed to be in place

There is NO pre-requisite for this subject.

7.5 Critical Cross-field Outcomes

CRITICAL CROSS-FIELD OUTCOMES	DESCRIPTION
CCFO1	Identify and solve problems with responses which display responsible decisions, using critical thinking, have been made.
CCFO2	Work effectively with others as a member of a team, group, organisation or community.
CCFO3	Organise and manage oneself and one's activities responsibly and effectively.
CCFO4	Collect, analyse, organise and critically evaluate information.
CCFO5	Communicate effectively, using visual, mathematical, and/or language skills in the modes of written and/or oral presentation.

8. STUDY UNITS/STUDY THEMES

8.1 Study Units

Refer to 7.2.2 Subject Breakdown Structure.

8.2 Assessment Opportunities and Assessment strategy

Refer to 6.2 Assessment Opportunities: Administration for the linkages. Furthermore, a comprehensive rubric will be published on Blackboard for each assignment and project work.

8.3 Subject Outcomes and Associated Assessment Criteria

Refer to 7.2.2 Subject Breakdown Structure for the linkages. Furthermore, a comprehensive rubric will be published on Blackboard for each assignment and project work.

8.4 Self-study Activities

End of chapter exercises and quizzes.

9. STUDENT ACKNOWLEDGEMENT

CAPE PENINSULA UNIVERSITY OF TECHNOLOGY
STUDENT ACKNOWLEDGEMENT
2021
DIPICTA: INFORMATION TECHNOLOGY
INFORMATION SYSTEMS II

ACKNOWLEDGE THAT YOU HAVE READ THE SUBJECT GUIDE AND UNDERSTAND ITS CONTENT

Please read the following statement and complete the spaces that follow. Make a copy of this page for your own reference. Hand-in this original page during the next class session.

I have read the subject guide of **INFORMATION SYSTEMS II** and understand the information and my responsibilities as a learner for this subject. I am also aware that the Subject Guide may be revised due to unforeseen circumstances and it remains my responsibility to adhere to the most recent guide.

Name & Surname:
(Print)

Student Number:

Contact number: (home/res) Contact number: (cell)

Signature:

Date: